The phone app: SSwinGG recorder

An app for an Android phone that connects to the bat sensor, records each shot between Start and Stop, and saves it as a CSV file with every reading of every axis. It is in the app folder.

It is a web app: it runs in Chrome and can be added to the home screen like any other app. Chrome on Android can talk to Bluetooth devices, so no app store and no Android Studio are needed.

What you need

- **An Android phone with Chrome.** iPhones cannot be used: Safari and every other iPhone browser lack Web Bluetooth.
- **The bat running stage 5 or stage 8.** The app reads the same 14-byte packet as the dashboard.
- **The computer that holds this kit**, to open the app the first time.

Open it on the phone

Browsers allow Bluetooth only on pages opened from `https` or from `localhost`. Pick one of these ways.

A. Over a USB cable

1. On the phone, switch on Developer options and USB debugging. Plug the phone into the computer and allow debugging when the phone asks.
2. On the computer, start a small web server in the app folder:

   ```
   cd app
   python3 -m http.server 8000
   ```

3. On the computer, open `chrome://inspect/#devices` in Chrome. Press **Port forwarding**, add port `8000` to `localhost:8000`, and tick **Enable port forwarding**.
4. On the phone, open `http://localhost:8000` in Chrome.

B. Over WiFi, without a cable

1. Start the web server as in A, step 2. Phone and computer must be on the same WiFi.
2. Find the computer's address. On a Mac: `ipconfig getifaddr en0`, for example `192.168.1.10`.
3. On the phone, open `chrome://flags` in Chrome, search for "Insecure origins treated as secure", enter `http://192.168.1.10:8000`, switch it to Enabled and press Relaunch.
4. Open `http://192.168.1.10:8000` on the phone.

C. From the internet. Once the app is published on an `https` address, for example with GitHub Pages, the phone simply opens that address. This has not been set up.

Then, in Chrome's menu, choose **Add to Home screen**. The app opens from its own icon after that.

Record a shot

1. Switch the bat on and wait for its slow blink.

2. Press **Connect sensor** and choose `SwingLab-Bat`. Allow Bluetooth, and Location or Nearby devices, if the phone asks.
3. Choose the shot type, or leave it on **Detect automatically**.
4. Press **Start**, play the shot, press **Stop**.
5. The file is saved at once to the phone's **Downloads** folder, and the list under **Recordings** shows it.

Keep the app on the screen while recording. It was not tested with the screen off or another app in front.

The files

Each shot is one file, named after its type and numbered: `pull_shot_001.csv`, `pull_shot_002.csv`, `drive_shot_001.csv`. A number is never used twice, even after a recording is deleted. Recordings made with the simulated bat start with `sim_`, so they cannot be mixed up with real ones.

One row per reading, about 50 a second, with these columns:

Column	Meaning
`sample_index`	0, 1, 2 and so on, in the order the readings arrived
`timestamp_iso`, `timestamp_ms`	When the phone received the reading, as a date and as milliseconds since 1970. The sensor sends no time of its own.
`elapsed_ms`	Milliseconds since Start was pressed
`ax_g`, `ay_g`, `az_g`	Acceleration in g
`gx_dps`, `gy_dps`, `gz_dps`	Rotation in degrees per second, with the still position already taken off by the bat
`accel_total_g`	Total acceleration in g
`speed_kmh`	Bat speed, by the dashboard's formula
`ax_raw` to `gz_raw`	The six numbers exactly as the sensor sent them, in counts
`temp_raw`	The sensor's temperature in counts. In degrees Celsius it is 23 + temp_raw / 512. Stage 5 sends 0.

The files open in Excel, Google Sheets, or Python with `pandas.read_csv`.

How the shot type is found

This is a first guess, not a tested classifier. No recordings labelled with their real shot type exist yet, so nothing could be trained or checked. The guess uses fixed rules:

What the recording shows	Guess
Fewer than 10 readings	unknown
Peak bat speed under 20 km/h	defensive shot
Faster, and turning to the leg side at the peak, rotation around z above +20 degrees per second	pull shot
Faster, and turning to the off side, below -20	cut shot
Faster, and neither	drive shot

The side rule is the dashboard's own, so it depends on the sensor being mounted the way stage 8 describes.

To build a real classifier, record labelled data first. Choose the shot type yourself before pressing Start, and record many shots of each type. The files are then named by their true type. The app still notes its guess in the list, so you can see how often the rules agree. With enough files, a model can be trained on them and put in place of `classify` in `app/recorder.js`.

Try it without a sensor

Press **Simulated bat**. It makes up a swing every three seconds, to the leg side, the off side and straight in turn. Everything else works as with a real sensor.

Limits

- **Android and Chrome only.** No iPhone.
- **CSV only.** Excel opens CSV files directly. Real `.xlsx` files are not made.
- **One folder.** Browsers save every file to Downloads, so the type is in the file name, not in a folder per type.
- **Gaps cannot be detected for certain.** The sensor numbers nothing, so a lost reading shows only as a longer step in `elapsed_ms`.
- **Not run with a real sensor yet.** It was tested in Chrome on a computer, with the simulated bat.